MTU15 thesis project — intuitive overview

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1. The question

Market structure (reform calendar in one paragraph).

Who actually sets the price (NOT a priori CCGT).

Why this matters.

Outcomes we care about, and why.

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2. Variation across hours: where granularity has economic content

Intuition.

Classifier.

What we see.

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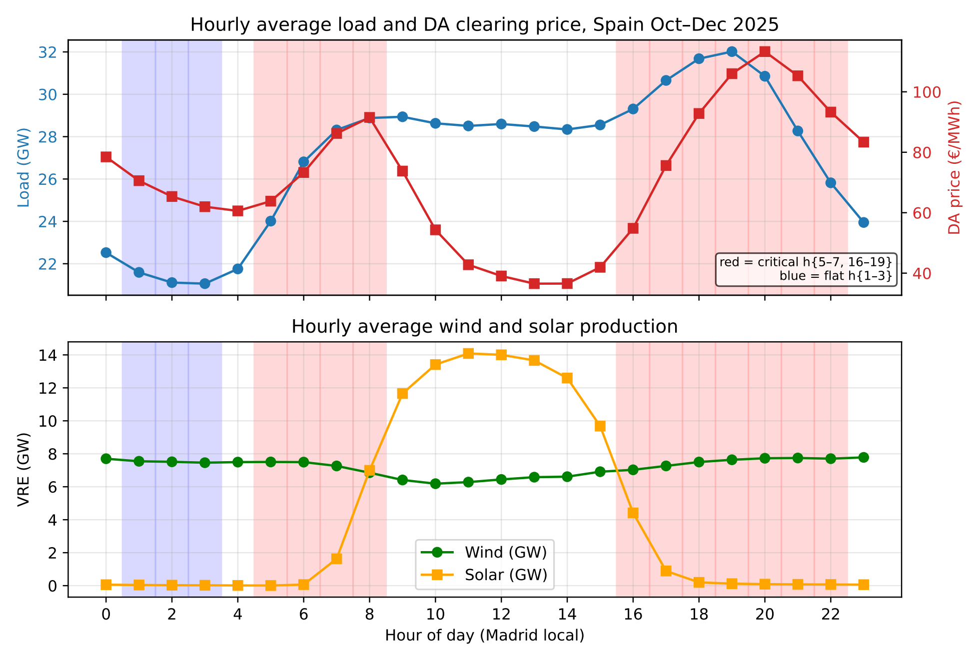

Figure 1:

Validation.

3. Variation across firms and technologies: how do CCGTs bid?

The object.

Metric: D_w.

Decomposition: Δp **vs** Δq.

What we see (post-MTU15-DA, critical hours, CCGT, Oct–Dec 2025).

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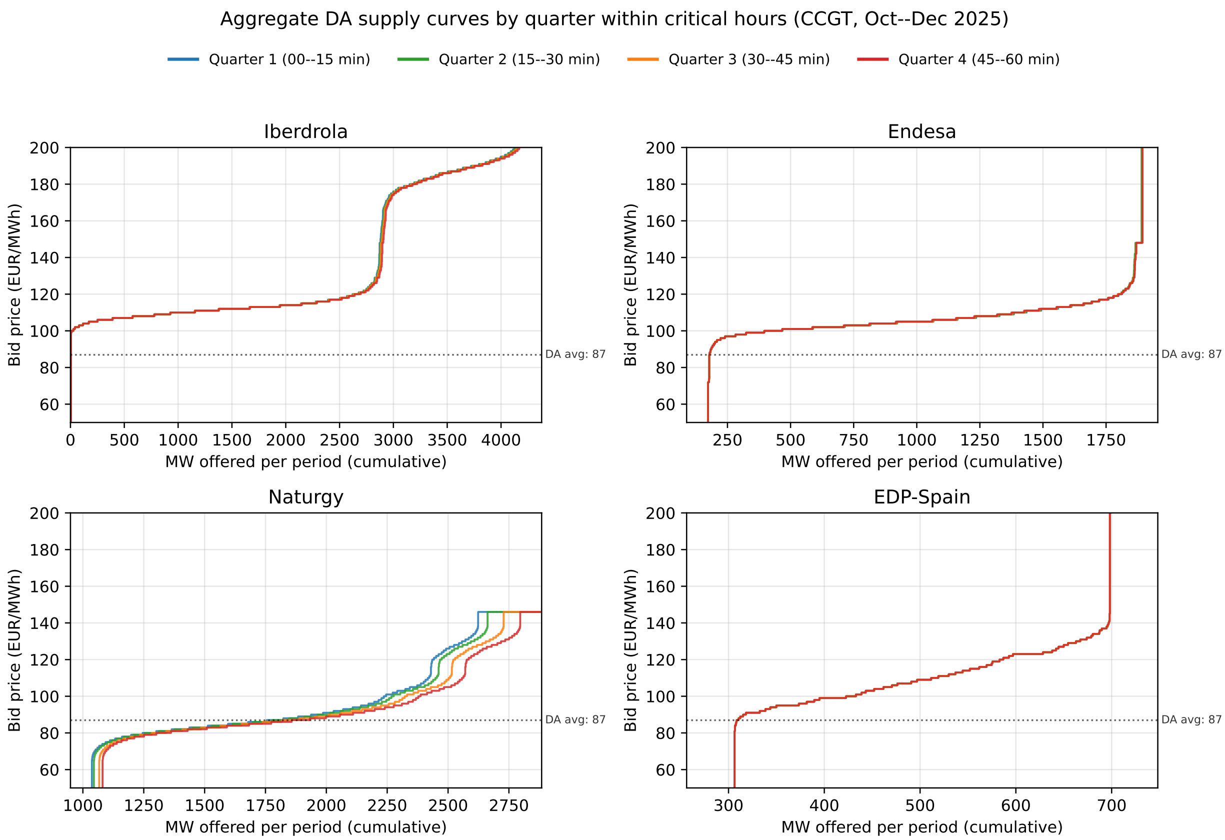

Figure 2:

Operational vs strategic decomposition.

4. The 2025 regulatory shock: blackout and reinforced operation

The blackout (28 April 2025).

What "reforzada" does in practice.

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What this lets us claim.

5. Reform windows we use, and outcomes we measure in each

Strategy in one sentence.

The reform calendar and our windows.

Reform	Date	Window and why

Outcomes we measure in each window.

Outcome	What it tells us / why it matters

6. MTU15 effect: critical/flat identification (ID15 and DA15)

Strategy. *(Gap to write: 3–4 sentences recapping the critical/flat identification — critical hours* $\{5, 6, 7, 8, 16, \dots, 22\}$ *are where within-hour residual demand varies and MTU15 has economic content; flat hours* $\{1, 2, 3\}$ *are the within-day control where MTU15 should not matter; the (critical − flat) differential across the reform date isolates the MTU15 effect from any day-level shock that hits both classes proportionally.)*

6.1 Specifications

Let u index a unit, d a date, p a market period (hour pre-MTU15, quarter post), h the clock-hour, s the IDA session, T_r the reform date, $\mathcal{C}$ the critical-hour set. All sell-side, in-band $|p_{\text{bid}} - \text{MCP}| \leq 140$ EUR/MWh.

Spec A — per-curve functional outcomes $Y \in \{\sigma_p, N_{\text{eff}}\}$, panel $(u, d, p[, s])$:

$$Y_{u,d,p[,s]} \; = \; \alpha + \alpha_u + \delta_d + \gamma_h + \beta \, \mathbf{1}\{d \geq T_r\} + \theta \, \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

α_u unit FE absorbed by within-unit demeaning; γ_h absorbs the crit-flat baseline; SEs clustered by date. θ is the DiD.

Spec B — aggregate clearing-price DiD, panel $(d, p[, s])$ over the OMIE clearing price P:

$$P_{d,p[,s]} \; = \; \alpha + \gamma_h + \beta \, \mathbf{1}\{d \geq T_r\} + \theta \, \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

Spec C — within-hour dispersion (post-only; pre has one period per hour, SD undefined). For each (unit, date, clock-hour [, session]) with all 4 quarters present:

$$D^{(u,d,h)}_{\text{price}} \; = \; \text{sd}_q\{\bar{p}_{u,d,h,q}\}\,, \qquad\qquad D^{(u,d,h)}_{\text{qty}} \; = \; \frac{\text{sd}_q\{m_{u,d,h,q}\}}{\text{mean}_q\{m_{u,d,h,q}\}},$$

where $\bar{p}_{u,d,h,q}$ is the unit's MW-weighted in-band mean price in quarter q and $m_{u,d,h,q}$ is its in-band MW. Regress on the critical indicator:

$$D^{(u,d,h)}_{\cdot} \; = \; \alpha + \alpha_u + \delta_d + \theta \, \mathbf{1}\{h \in \mathcal{C}\} + \varepsilon$$

System-level analog drops the unit dimension and uses the clearing-price SD across the 4 quarters.

6.2 Sample windows (reforzada handled by tight pre/post)

Reform	Pre window	Post window	Reforzada
MTU15-IDA	2024-12-19 → 2025-03-18 (3 mo, ISP15)	**2025-03-19** → **2025-04-27** (40 d, clean)	absent both arms
MTU15-DA	2025-07-01 → 2025-09-30 (3 mo)	2025-10-01 → 2025-12-31 (3 mo)	active both arms, level constant

For MTU15-IDA the post-window is pre-blackout (reforzada absent throughout) so θ identifies the MTU15-IDA effect cleanly. For MTU15-DA both arms sit under continuous reforzada (active since 2025-04-28); the DiD critical-flat interaction strips the level shift as long as the reforzada critical-vs-flat differential is stable across the cutover.

6.3 Spec A — per-curve σ_p and N_{eff}

Reform	Outcome	Pre crit	Post crit	Pre flat	Post flat	DiD θ	SE	t
MTU15-IDA	σ_p	8.78	7.52	11.53	8.84	+1.71	0.34	5.06
MTU15-IDA	N_{eff}	3.98	5.07	5.97	7.31	+0.10	0.12	0.83
MTU15-DA	σ_p	19.13	21.06	16.34	16.89	+1.36	0.34	4.07
MTU15-DA	N_{eff}	3.22	3.58	3.06	3.28	+0.14	0.04	3.96

Reading. *(Gap. Key points: σ_p DiD positive and significant for BOTH reforms — MTU15 unlocks a wider in-band price ladder in critical hours, the strategic margin granularity creates. N_{eff} DiD: significant only for DA15, suggesting quantity-side fragmentation responds to the DA reform but not detectably to the IDA reform within the 40-day clean window.)*

6.4 Spec B — aggregate clearing-price DiD

Reform	Spec	Pre crit	Post crit	Pre flat	Post flat	DiD θ (EUR/MWh)	SE	t
MTU15-IDA	B0 (no FE)	114.0	38.3	84.2	19.8	−11.32	3.72	−3.04
MTU15-IDA	B1 (date FE)	—	—	—	—	−11.18	3.75	−2.98
MTU15-DA	B0 (no FE)	79.4	86.9	85.2	66.0	+26.70	2.51	+10.65
MTU15-DA	B1 (date FE)	—	—	—	—	+26.70	2.54	+10.50

Seasonality. The pre/post arms span large seasonal transitions (ID15 winter → early spring; DA15 summer → autumn) and the level changes are correspondingly large. Spec B1 absorbs every common date-level shock via date FE; the DiD is essentially unchanged (−11.32 → −11.18 ID15; +26.70 → +26.70 DA15), so the headline numbers are *not* driven by simple date-level seasonality. *(Residual concern, to flag in the prose: a differential seasonal* trend *in the (crit − flat) differential itself — which date FE does not catch — would still confound the DiD. The standard same-calendar-month robustness against prior years is not directly feasible in the tight windows used here; the cleanest available fallback is the Spec C cross-sectional within-hour dispersion test below, which is post-only and so immune by construction to pre/post seasonal drift.)*

Reading. *(Gap. The headline empirical fact: opposite signs. ID15 — critical-hour IDA prices fell EUR 11/MWh more than flat-hour ones after MTU15-IDA (granular intraday absorbed within-hour scarcity, compressing the critical peak). DA15 — critical-hour DA prices rose EUR 27/MWh more than flat (granular day-ahead pricing extracted higher rents in scarce quarters; within-hour ladder now realised in the clearing price).)*

6.5 Spec C — within-hour dispersion (post-MTU15 only)

Reform	Object	Outcome	Mean flat	Mean crit	θ_{crit}	SE	t
MTU15-IDA	per-unit	D_{price} (EUR/MWh)	0.12	0.50	+0.38	0.05	8.02
MTU15-IDA	per-unit	D_{qty} (CV)	0.007	0.031	+0.023	0.003	8.96
MTU15-IDA	system	SD_{price} (EUR/MWh)	2.50	7.04	+4.55	0.40	11.46
MTU15-DA	per-unit	D_{price} (EUR/MWh)	0.21	1.53	+1.32	0.11	12.10
MTU15-DA	per-unit	D_{qty} (CV)	0.008	0.041	+0.032	0.002	18.77
MTU15-DA	system	SD_{price} (EUR/MWh)	2.49	6.46	+3.97	0.37	10.72

Reading. *(Gap. The falsification: every within-hour dispersion outcome is* 3–$7\times$ *larger in critical than flat hours post-MTU15. This is the direct confirmation that MTU15 unlocks within-hour discrimination only where within-hour residual demand actually varies. The system-level* $SD_{price} \sim EUR\,6$–$7/MWh$ *in critical hours vs* $\sim EUR\,2.5$ *in flat is the realised cross-quarter clearing-price spread — the empirical face of the unlocked discrimination margin.)*

6.6 Robustness (to add)

(Gap to write. Planned: extended pre windows (full pre-MTU15-IDA back to 2024-06-14; full pre-MTU15-DA back to 2024-06-14) for sample-length sensitivity; same-calendar-month restriction; per-firm and per-tech breakdowns of the DiD (trivial off the existing panel); for DA15, acknowledgement that no clean placebo exists for separating MTU15-DA from reforzada beyond the critical-flat differencing.)